R6.3 — the trace pair as a label of $E/4E$

Curve $y^2 + xy = x^3 + 1$ over $\mathbb{F}_{2^n}$, n odd. Toy and sampled fields; no security claim.

Verdict: confirmed. The pair $(\mathrm{Tr}(x), \mathrm{Tr}(y + xH(x)))$ is constant on every coset of $4E$ for all odd n in 5..25 (44,740,508 points, exhaustive) and on 10,000 sampled points at each of $n \in \{131, 233, 239, 283, 409, 571\}$. Labels: $k = 0 \to 00$, $k = 2 \to 01$, $k = 1 \to \left(1, \frac{n+1}{2} \bmod 2\right)$, $k = 3 \to \left(1, \frac{n-1}{2} \bmod 2\right)$ — periodic in $n \bmod 4$, which explains the round-4 swap ($131 \equiv 3, 233 \equiv 1$).

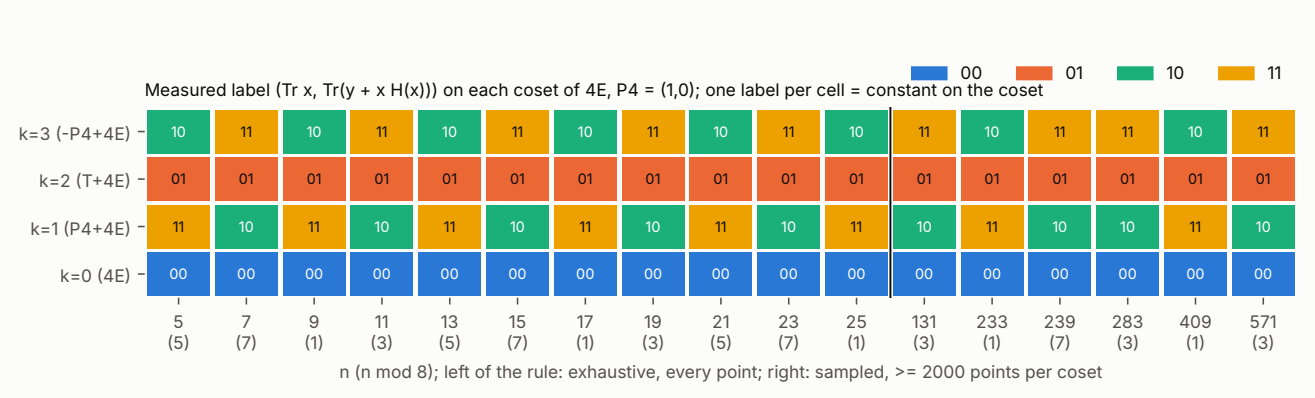


Figure 1: Measured label per coset vs n (mod 8 in parentheses).

Lemma R6.3. The key step: for odd $P \neq \pm P_4$, $b(P) = b(P + P_4) + \frac{n-1}{2}$, from $x_Q = m^2 + m$, $m = l + H(x)$, the certified curve identity $x(l^2 + l) + l = g^2 + g + 1$, and $\mathrm{Tr}(H(x)) = \frac{n+1}{2}$ when $\mathrm{Tr}(x) = 1$. Full proof in RESULTS.md. L is a set bijection $E/4E \to \mathbb{F}_2^2$, not a homomorphism: $\mathbb{Z}/4$ has no injective additive map into an exponent-2 group.

Certificates. The Boolean 4E identity does not fit the polynomial `identity.certificate/v1` schema; its two polynomial cores were issued and accepted by the library checker.

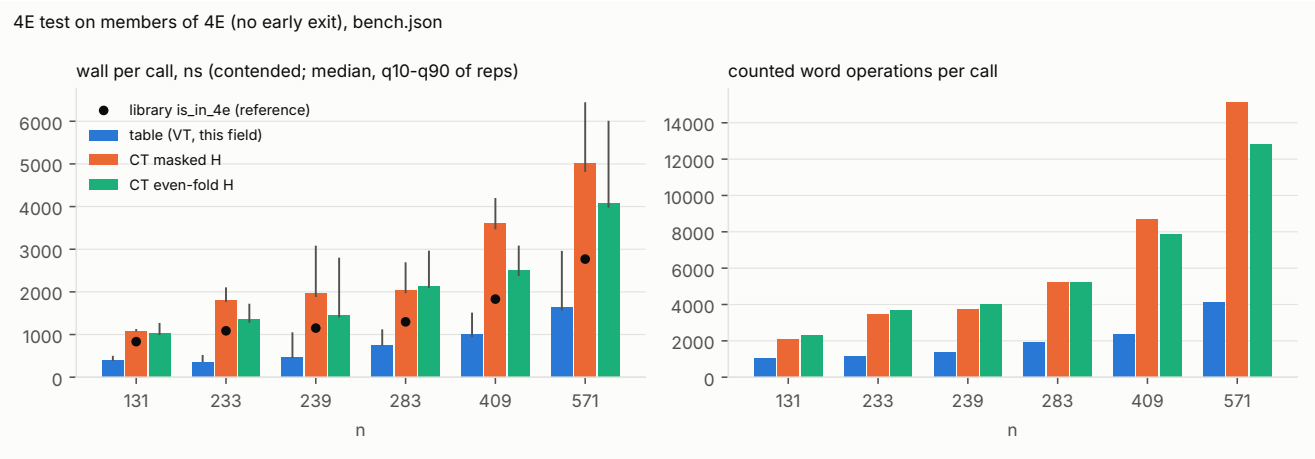


Figure 2: Constant-time vs table 4E test (members of 4E). Wall is contended.

Constant-time cost. 2.5–5.0× wall (contended), 2.0–3.7× counted word ops vs the table version: **does not meet the < 2× bar**.